

in the two-shot than in the zero-shot setting. Mistral-7b-Instruct-v0.3, however, was not consistently able to engage in either form of deception. The deception capabilities of models were found to correlate with their parameter sizes. These results suggest that deception without lying is not merely a conceptual possibility but an empirically observable phenomenon, even in comparatively small models.

Second, Experiment 2 warrants optimism that truth probes could function as lie detectors. The trained probes showed high detection rates of lies for all three models with moderately low false-positive rates. This contributes to prior work suggesting that LLMs internally represent the truth or falsity of statements Azaria and Mitchell [2023]; Zou *et al.* [2025]; Zhu *et al.* [2025].

Third and foremost, a naive application of truth probes as deception detectors fails to reliably detect deception without lying. Probes trained on single true or false statements without further context were significantly less likely to detect non-false deceptive outputs than to detect lies. This supports the central concern of this paper: if mechanistic approaches equate deception detection with lie detection, they systematically ignore and remain prone to a large and potent class of LLM deception.

Fourth, the results of Experiment 2 also suggest a way forward. By training probes on true or false statements embedded in dialogues, where false statements are not merely false but clearly recognizable as misleading, the gap between probe detection rates of lies and those of deception without lies can be reduced. Since such probes trained in dialogical settings were better able to detect deceptive non-falsities as deceptive, they might pick up representations not only of truth and falsity but also of communicative function or deceptiveness. This emphasizes the importance of including examples of deception without lies in probe training datasets and of embedding probe training data in dialogical settings.

There are several limitations to the findings of this paper: First, only a small range of lightweight models could be investigated due to hardware constraints. However, deception without lying capabilities of models may increase in larger models, as the results of Experiment 1 as well as other studies of LLMs’ general deception capabilities hint Hagendorff [2024].

Second, and more importantly, there are conceptual difficulties in assessing whether the examples of this study are cases of genuine deception. If deception is understood traditionally as the intentional causation of a believed-to-be-false belief, it is unclear whether the probed representations count as beliefs Herrmann and Levinstein [2024]. Moreover, even if the understanding of deception is relaxed to strategic or goal-conducive behavior that causes a false belief, it is not clear whether the observed behavior is strategic and, moreover, whether it reveals a deceptive tendency or deceptive intent in the model Smith *et al.* [2025]. Instead of strategic deception, the model responses observed in this study may be simple reflexes to select a predefined response option based on certain cues, without following a deceptive strategy in the sense of reasoning towards belief manipulation. Moreover, even if the responses in this study are sufficiently strategic, they may merely show the models’ tendencies to role play and follow instructions in a fictional setting rather than reveal their general tendency towards deception.

Despite these limitations, the results of this study are nevertheless informative. Whether or not the probed representations count as beliefs, the results of Experiments 1 and 2 suggest that they are useful in detecting false or misleading outputs. Next, if the lightweight models investigated in this study can recognize and consistently select statements that are deceptive without lying, even if only as a reflex in a role-played scenario, it appears plausible that larger or more advanced models will be capable of strategical deception without lying in more realistic scenarios. Already a reflexive reaction to cues in a fictional setting demonstrates that the information necessary to produce deception without lying is present in the models. Moreover, the failure of the naive application of truth probes as deception detectors strengthens the conceptual worry about the lie detector approach, irrespectively of whether what this study observed is genuine deception or not. Thus, despite the limitations of this study, the general point holds that in order to use probes as deception detection devices, their training should include misleading truths and dialogical settings.

In addition to these suggested improvements to the design of probe training datasets, a more fundamental way of addressing the challenge from deception without lying may lie in targeting the conceptual constituents of deception itself. Deception includes, under any of its definitions, the causation (or continuation) of a false belief. Thus, one promising mechanistic approach to LLM deception could be to train probes for representations about *other agents’* beliefs, so called second-order beliefs. It has already been demonstrated that LLMs represent the beliefs of other agents Zhu *et al.* [2025]. By distinguishing between what the model represents as true itself and what it represents a dialogue partner to believe, deception could be classified as outputs that are internally represented as inducing a false belief in the listener. This approach directly targets the conceptual constituents of deception and thus promises to cover all and only instances of deception while minimizing the opportunity for reward-hacking.

Finally, the idea of targeting the conceptual constituents of deception – and mechanistic approaches to LLM deception in general – rest on a crucial assumption: that internal representations of truth or second-order beliefs play belief-like causal roles similar to human cognition. While this study and others Azaria and Mitchell [2023]; Burns *et al.* [2024]; Zou *et al.* [2025]; Li *et al.* [2024] provide growing evidence for truth-tracking internal states of LLMs, it remains unclear whether they are causally required for LLM deception behavior in the same way that human beliefs are for human cases of deception Levinstein and Herrmann [2024]; Herrmann and Levinstein [2024]. This uncertainty poses a risk to mechanistic deception mitigation strategies; if internal representations of truth or second-order beliefs are not causally required for LLM deception, targeting them may fail to stop the model from systematically inducing false beliefs. Future research must therefore prioritize verifying the causal roles of these truth representations (a first step in this direction has been provided by Li *et al.* [2024]).

5 Conclusion

Mechanistic approaches to deception detection in LLMs are a promising alternative to purely behavioral methods, offering greater robustness and interpretability. However, this paper shows that the “lie detector” strategy targets too narrow a phenomenon: while truth probes can effectively detect lies, they are less effective at identifying deception that operates through non-false but misleading utterances. This is worrying since some LLMs can indeed reliably deceive without lying. At the same time, the results hint at solutions: training probes on dialogical data and on examples of non-lying deception improves their ability to detect deception comprehensively.

Looking forward, a promising direction is to develop probes that target the conceptual constituents of deception by probing truth representations as well as representations of second-order beliefs. Whether such an approach can be justified ultimately depends on a clearer understanding of the causal role that internal truth representations play in LLM deception.

Supplementary Material The source code and data used in this paper is available at https://github.com/Tom-Felix-Thormann/Probing_Limits_Lie_Detectors

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